

Deployment-Aware Model Selection and Adaptation for Few-Shot Predictive-Model Instantiation in Digital Twin Platforms

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ABSTRACT

Digital twin (DT) platforms for computing networks can reuse pretrained predictive models from existing centers to reduce the computation required for center-specific model instantiation. In this paper, instantiation refers to the predictive-model component of a DT rather than the complete DT lifecycle. However, a new or reconfigured center may exhibit substantial differences in resource configurations and workload characteristics, making a fixed architecture suboptimal for the target environment. At the target center, only a small support set is available for parameter adaptation and a small validation set for model selection. Accurate predictive-model instantiation therefore requires selecting a suitable architecture and adapting its parameters from such limited target data while ensuring that the selected model satisfies deployment limits. Existing parameter-adaptation-based few-shot methods typically use predefined architectures and rarely account for deployment limits, limiting their applicability to heterogeneous computing environments. To address this challenge, we propose Model Selection and Adaptation for Digital Twin Instantiation (MSA-DTI), a few-shot predictive-model instantiation framework that integrates target-specific architecture selection and parameter adaptation under deployment limits. MSA-DTI first constructs a candidate model bank by pretraining diverse architectures on historical workload data, filters out infeasible candidates according to the deployment limits, and adapts the remaining candidates using limited target samples. To make model selection more conservative given this small validation set, it compares the best-performing adapted alternative with a fixed adapted reference model and uses a preset relative reduction in validation loss as the replacement criterion. On the controlled synthetic heterogeneous-center benchmark, MSA-DTI achieves the lowest mean squared error (MSE) among the compared methods, reducing MSE by 14.60% on average across paired cases relative to the pretraining-and-fine-tuning baseline while satisfying the deployment limits across all main evaluation cases.

1. Introduction

A shared digital twin (DT) platform maintains center-specific predictive models for monitoring, prediction, analysis, and decision making [1–3]. These models transform synchronized monitoring histories into workload forecasts that can support resource-management decisions. The present work addresses this predictive-model component; sensing, state synchronization, simulation, and actuation remain services of the surrounding DT platform. To reduce the computation required for model instantiation, the platform can reuse models pretrained using data from existing centers. However, a new or reconfigured center may differ substantially from existing centers in resource configurations, monitored-variable availability, workload patterns, and data distributions, making a fixed pretrained architecture suboptimal for the target environment. Concurrent model maintenance also creates computation and storage pressure, so the platform sets two per-model deployment limits: estimated operation count per inference and parameter count. The operation count describes model-side inference computation, while the parameter count describes model size. In this setting, the target center has only limited historical observations, available as a small support set for parameter adaptation and a small validation set for model selection. The limited support set defines the few-shot target-adaptation setting considered in this work. Selecting a suitable architecture and adapting its parameters from such limited target data are challenging, while the resulting DT model must satisfy both deployment limits. Figure 1 shows this setting.

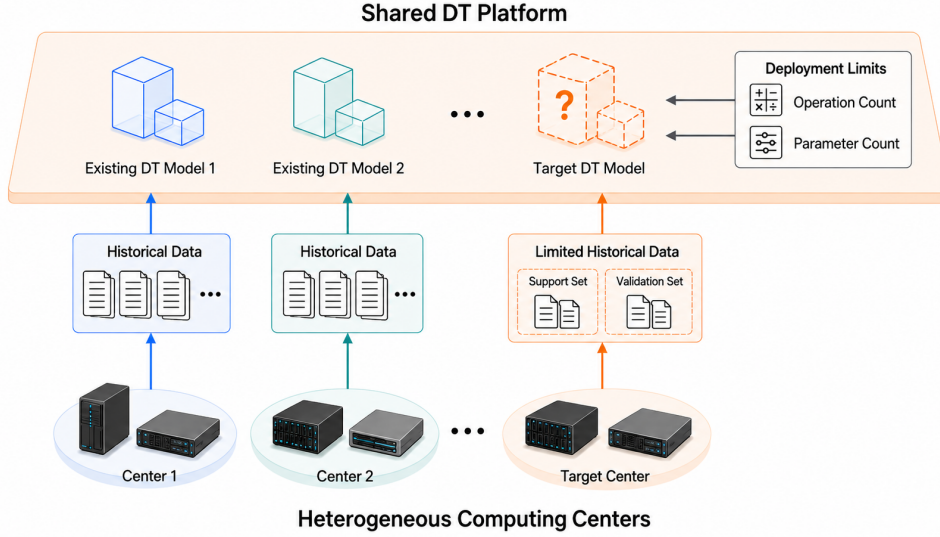


Figure 1: System setting for few-shot digital twin model instantiation with limited target data and deployment limits.

Existing parameter-adaptation-based few-shot and meta-learning methods adapt model parameters from limited data [9, 10], and few-shot adaptation has also been applied to DT modeling [11]. These methods usually keep the model architecture fixed and do not explicitly incorporate per-model deployment limits into target-side adaptation, which limits their applicability to heterogeneous target centers. Deployment-aware architecture selection considers model performance and deployment cost [12, 13, 26], while meta-learning has also been used to adapt architectures and fine-tuning strategies to new few-shot tasks [30–32]. Such methods learn or search an architecture or adaptation structure for the target task, and H-Meta-NAS further considers hardware-constrained architecture adaptation [30]. Unlike these search-based settings, the target-side task here starts from the pretrained models already available at the DT platform. It therefore concerns selection and adaptation of reusable models under deployment limits rather than target-side architecture or adaptation-structure search.

To address this problem, we propose Model Selection and Adaptation for Digital Twin Instantiation (MSA-DTI). MSA-DTI first builds a candidate bank, defined as a set of models pretrained on historical data from existing centers, and removes candidates that violate the target deployment limits. It then adapts all feasible candidates using the same support set and update budget. Because the target validation set is small, MSA-DTI compares the best-performing adapted alternative with a fixed adapted reference model and uses a preset relative reduction in validation loss as the replacement criterion. This design enables target-specific architecture selection without target-side architecture search and prevents replacement based only on marginal validation-loss reductions. The selected architecture and adapted parameters form the predictive-model instance used by the target center’s DT.

The target decision set is a finite collection of heterogeneous architecture–initialization pairs after common target adaptation. Related model-zoo and model-hub methods already rank, tune, aggregate, or select pretrained models. The narrower distinction here is a deployment-filtered bank of separately source-trained full models, followed by equal-budget target adaptation and a protected-reference replacement rule calibrated on data disjoint from final evaluation. This also differs from PT+FT, which adapts only one fixed reference architecture, and from NAS and meta-NAS, which search or adapt an architecture in the target stage. The relative margin is a separate final-selection rule: it does not construct the bank, change the target updates, or create the gain from cross-architecture adaptation.

This work makes the following systems and empirical contributions.

- **Deployment-aware DT instantiation problem.** We formulate the predictive-model instantiation task at a target center as selection and adaptation from limited target data, subject to two explicit per-model deployment limits and strict separation of adaptation, validation, and final evaluation data.
- **Integrated instantiation protocol.** We develop MSA-DTI as a reproducible platform protocol that filters a prespecified reusable model bank, adapts every feasible candidate under a common target budget, and

exhaustively evaluates the finite feasible set. This is an integrated systems design rather than a new general-purpose joint optimization solver.

- **Protected-reference decision rule and evidence.** We instantiate a relative-margin rule calibrated on a disjoint pool and evaluate the complete protocol on held-out heterogeneous-center cases, source-sensitivity studies, and an Alibaba workload-domain transfer audit. The results report both gains and failure boundaries, including sample-sensitive candidate discovery and unsuccessful external-domain margin recalibration.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 defines the system model and problem. Section 4 presents MSA-DTI. Section 5 reports the experiments, and Section 6 concludes the paper.

2. Related Work

2.1. Digital Twin Modeling and Platform Deployment

Digital twin research uses virtual models for monitoring, prediction, simulation, and control [1–4]. In communication and computing systems, digital twins also represent system states and services and support model updates [5–8]. Platform studies further consider service orchestration and computation offloading under resource constraints [27, 28]. DT-specific studies have formulated server-side model selection under synchronization and execution constraints [40], learned contextual AI-model selection mappings with digital-twin assistance [41], and selected forecasting models according to time-series characteristics and available computing resources [42]. These studies establish model selection as a DT systems problem, but they do not address the present target-side protocol: equal-budget few-shot adaptation of a finite bank of separately source-trained workload-forecasting models, hard filtering before adaptation, and protected-reference replacement from a small validation set.

2.2. Few-Shot Learning and Meta-Learning

Few-shot learning and meta-learning reuse prior knowledge to adapt models from limited data [9, 10, 24]. Recent methods improve cross-task and cross-domain adaptation through context modeling and transferable representations [20–23], and limited-data learning has also been applied to DT construction and workload prediction [11, 25]. Most parameter-adaptation methods keep the architecture fixed and mainly adapt model parameters, without explicitly accounting for per-model deployment limits, which limits their applicability to heterogeneous target centers.

2.3. Pretrained Model-Zoo Selection and Adaptation

Model-zoo and model-hub research has established that selecting and adapting among pretrained models is itself an existing problem. Zoo-Tuning adaptively transfers and aggregates parameters from multiple source models [33]; ranking-and-tuning methods rank heterogeneous pretrained models before fine-tuning or use several ranked models to tune a target model [34]; and model soups select and average fine-tuned candidates [35]. Zero-shot AutoML selects pretrained-model and fine-tuning pipelines from dataset meta-features [38], Quick-Tune jointly searches pretrained models and fine-tuning hyperparameters under a time budget [37], and ZooKT transfers knowledge from a model zoo for few-shot learning [39]. Transfer-learning theory also studies the limits of model selection when source–target relationships vary across hypothesis classes [36].

MSA-DTI does not claim general novelty for model selection or fine-tuning. Its JNCA-oriented systems contribution is a deployment-aware instantiation protocol for networked DT platforms. The protocol operates on a finite heterogeneous bank of separately source-trained full workload-forecasting models: hard deployment limits are applied before adaptation, every feasible candidate receives the same target update budget, and replacement of a protected reference requires a relative margin calibrated on a disjoint pool. The method returns one feasible model instance rather than aggregating model parameters. Candidate discovery is sample-sensitive, so the contribution is the evaluated instantiation design and its explicit safety boundary, not a claim that the screening rule uniquely recovers an architecture set.

2.4. Deployment-Aware Architecture Selection

Deployment-aware neural architecture search evaluates model performance together with platform or hardware cost [12, 13, 26]. Few-shot, zero-shot, and training-free NAS methods reduce architecture evaluation or search cost [14–16].

In Ref. [14], few-shot refers to the use of several sub-supernets for architecture evaluation rather than adaptation from a small target support set.

Meta-learning has also been combined with architecture search. T-NAS learns a transferable meta-architecture that is adapted to new tasks [31], while H-Meta-NAS extends task-adaptive architecture search to hardware-constrained deployments [30]. Neural fine-tuning search instead starts from a pretrained backbone and searches the adaptation structure, including which components to fine-tune or augment [32]. These studies show that architecture decisions and few-shot adaptation can be coupled.

MSA-DTI addresses a different target-side protocol. Unlike methods that learn or search a target architecture or adaptation structure, it operates over a finite bank of separately source-trained full-model candidates. Deployment feasibility is determined before target adaptation, whereas candidate preference is evaluated only after the feasible candidates have been adapted with the limited target support set. The resulting problem is therefore target-side instantiation over reusable frozen models under deployment limits rather than target-side architecture or adaptation-structure search.

3. System Model and Problem Formulation

3.1. System Setting and Target Model

We consider a digital twin platform that maintains center-specific DT models for heterogeneous computing centers. Let C_{src} denote the existing centers, and let D_{src} denote their combined historical workload data. In the experiments, each architecture-specific source initialization is trained on this same pooled source-center data. We use $C1$ only as the identifier for this legacy source-pooled initialization bank; thus C1-A57 is the C1 initialization for architecture index A57, not a center identifier or an additional data split.

For each instantiation case, let c denote a new or reconfigured target center. Let D_c^{sup} and D_c^{val} denote its support and validation sets introduced in Section 1. All centers are mapped to a common platform-wide metric schema. A center may observe only a subset of the metric positions, but the input dimension remains fixed after unavailable positions are masked.

Let $\mathcal{A}$ denote a predefined architecture space. For $a \in \mathcal{A}$, let $f_{a,\theta}$ denote a model with parameter vector θ . The predictive-model component instantiated for the DT of center c is

$$\text{DT}_c = (a_c, \theta_c), \quad (1)$$

where a_c is the selected architecture and θ_c is its adapted parameter vector. We retain the notation DT_c as shorthand for this component; it does not represent the complete DT system. Model instantiation must therefore determine both terms.

3.2. Candidate Models and Deployment Limits

A candidate model is written as $q = (a_q, \theta_q^0)$, where $a_q \in \mathcal{A}$ is its architecture and θ_q^0 is its parameter vector before target adaptation. Let $\mathcal{Q}$ denote a finite set of candidates, called the candidate bank.

For target center c , let B_c^F and B_c^P denote the operation-count and parameter-count limits, respectively. Let $F_c(a_q)$ be the estimated operation count of architecture a_q for one inference, and let $P_c(a_q)$ be its parameter count. The same counting rule is used for all candidates; the complete rule is given in Supplementary Section S1.

The candidates satisfying both limits form

$$\mathcal{Q}_c^{\text{fea}} = \{q \in \mathcal{Q} \mid F_c(a_q) \leq B_c^F, P_c(a_q) \leq B_c^P\}. \quad (2)$$

Here, $\mathcal{Q}_c^{\text{fea}}$ is the feasible candidate set. Both deployment limits are hard constraints rather than penalty terms in the task loss. The two counts are deterministic model-level deployment indicators rather than direct guarantees of hardware latency or memory use. Their relation to measured latency and model size is evaluated in Supplementary Section S5.

MSA-DTI uses a fixed reference candidate q^{ref} as the fallback against which alternative candidates are evaluated. The reference is fixed offline and is not re-selected using target data. We define a target limit setting as supported when $q^{\text{ref}} \in \mathcal{Q}_c^{\text{fea}}$. If the reference candidate violates an assigned limit, MSA-DTI reports UNSUPPORTEDLIMIT rather than changing the reference after observing target validation data. This preserves the fixed reference used by the calibrated selection rule.

3.3. Few-Shot Model Instantiation Problem

Let $\mathcal{L}(\mathcal{D}; a, \theta)$ denote the mean task loss of model $f_{a, \theta}$ on dataset $\mathcal{D}$. Let T denote the number of target updates. For a feasible candidate q , let $\theta_{c, q}^{(T)}$ denote its parameters after T updates on $\mathcal{D}_c^{\text{sup}}$. Its validation loss after adaptation is

$$J_c(q) = \mathcal{L}(\mathcal{D}_c^{\text{val}}; a_q, \theta_{c, q}^{(T)}). \quad (3)$$

For a supported target limit setting, the few-shot model instantiation problem is to determine the target DT instance from $\mathcal{Q}_c^{\text{fea}}$ using only $\mathcal{D}_c^{\text{sup}}$ and $\mathcal{D}_c^{\text{val}}$. This requires selecting a suitable architecture and adapting its parameters from limited target data while the returned model remains within both deployment limits. The selected candidate determines a_c and θ_c in Eq. (1).

We solve this problem using MSA-DTI, whose target-side procedure consists of fixed-budget parameter adaptation, exhaustive evaluation over the finite feasible candidate set, and reference-based final selection.

4. The Proposed MSA-DTI Method

4.1. Method Overview

Model Selection and Adaptation for Digital Twin Instantiation (MSA-DTI) has three stages. Offline preparation screens the architecture space on architecture-screening cases, constructs the candidate bank, fixes the reference candidate, and calibrates the relative margin on a disjoint margin-calibration pool. Deployment filtering removes candidates that do not satisfy the target limits. Target adaptation and selection then adapt the remaining candidates and select the final model.

Figure 2 shows the overall three-stage workflow of MSA-DTI.

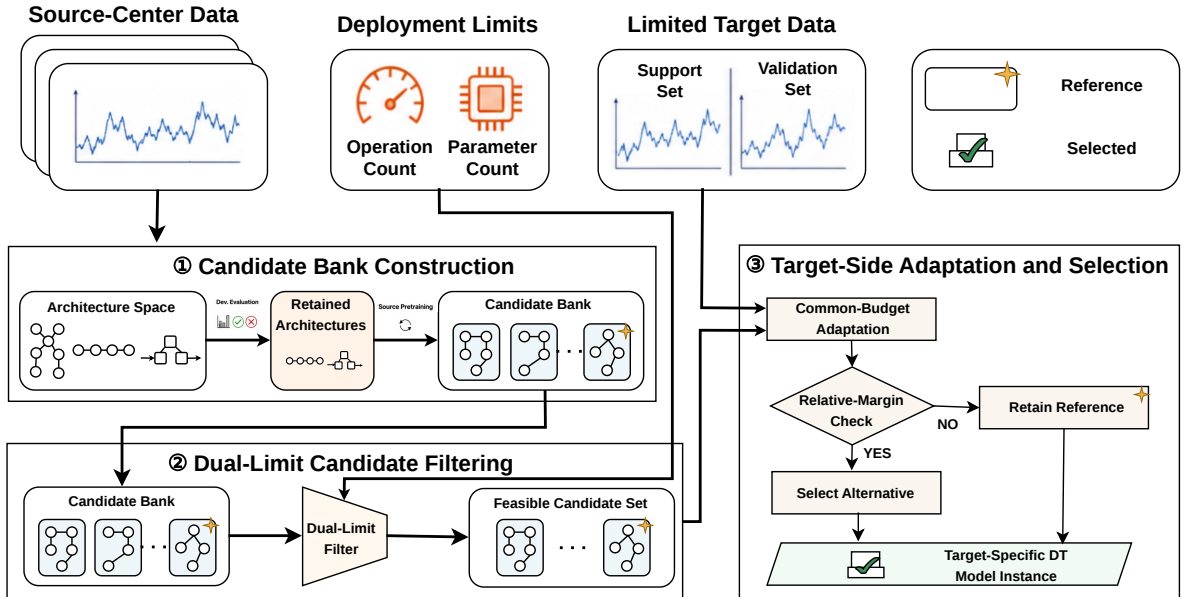


Figure 2: Overall three-stage workflow of MSA-DTI.

4.2. Offline Candidate Bank Construction

MSA-DTI screens the architecture space on architecture-screening cases that are separate from margin calibration and final target evaluation. Each screening case contains support, validation, and check sets, and the validation and check losses are used by the preset screening rule.

A reference architecture $a^{\text{ref}} \in \mathcal{A}$ is fixed before screening and always retained. Alternative architectures are retained according to the preset screening rule. Each retained architecture is then trained on $\mathcal{D}_{\text{src}}$, and each architecture-initialization pair forms a candidate $q = (a_q, \theta_q^0)$ in $\mathcal{Q}$. A retained architecture may therefore contribute more than

one candidate when multiple frozen source initializations are kept. One candidate with architecture a^{ref} is fixed as the reference candidate q^{ref} . In Fig. 3, $a^{(1)}, a^{(2)}, \dots$ denote example architectures in $\mathcal{A}$, $a_1, \dots, a_M$ denote retained architectures, and $q_1, q_2, \dots$ denote candidates in $\mathcal{Q}$, with q^{ref} denoting the fixed reference candidate.

The bank and reference candidate are fixed before relative-margin calibration and final target evaluation. Figure 3 summarizes this process. The exact screening and source-training settings are reported in Section 5.1 and Supplementary Section S2.

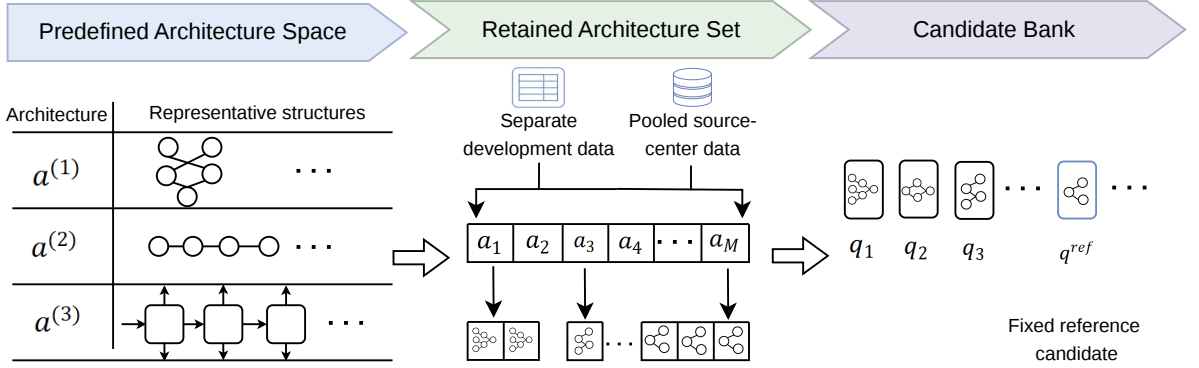


Figure 3: Offline construction of the candidate bank and reference candidate.

4.3. Deployment Filtering

For target center c , MSA-DTI forms $\mathcal{Q}_c^{\text{fea}}$ using Eq. (2) before target adaptation. Operation count and parameter count do not change during parameter adaptation, so candidates outside this set can be removed before target updates. If $q^{\text{ref}} \notin \mathcal{Q}_c^{\text{fea}}$, MSA-DTI returns UNSUPPORTEDLIMIT.

4.4. Target Adaptation

Each feasible candidate starts from θ_q^0 . Let t denote the update index, and let $\eta_t > 0$ denote the step size at update t . Candidate q is adapted by

$$\begin{aligned} \theta_{c,q}^{(0)} &= \theta_q^0, \\ \theta_{c,q}^{(t+1)} &= \theta_{c,q}^{(t)} - \eta_t \nabla_{\theta} \mathcal{L} \left(D_c^{\text{sup}}; a_q, \theta_{c,q}^{(t)} \right), \\ t &= 0, \dots, T-1. \end{aligned} \quad (4)$$

Here, ∇_{θ} denotes the parameter gradient. The architecture remains fixed during adaptation. All feasible candidates use the same support set and update rule, and their validation losses are then computed using Eq. (3). In the experiments, these gradient updates are implemented using stochastic gradient descent (SGD), as specified in Section 5.1.

4.5. Reference-Based Selection

Because the validation set is small, MSA-DTI uses a preset relative margin so that a marginal post-adaptation loss reduction does not by itself trigger replacement of the fixed reference.

Let $\mathcal{Q}_c^{\text{alt}} = \mathcal{Q}_c^{\text{fea}} \setminus \{q^{\text{ref}}\}$ denote the feasible alternatives. Because this set is finite, MSA-DTI exhaustively evaluates all adapted feasible alternatives. If this set is empty, MSA-DTI returns the adapted reference candidate. Otherwise, let $\tilde{q}_c \in \arg \min_{q \in \mathcal{Q}_c^{\text{alt}}} J_c(q)$ denote an alternative with the lowest validation loss. Exact ties at the stored numerical precision are resolved deterministically by lower parameter count, then lower estimated operation count, then lower architecture index, and finally the fixed source-candidate token. This lexicographic rule affects only ties among alternatives.

Let $\tau \in [0, 1)$ denote a preset relative margin. The final candidate is

$$q_c^* = \begin{cases} \tilde{q}_c, & J_c(\tilde{q}_c) \leq (1 - \tau) J_c(q^{\text{ref}}), \\ q^{\text{ref}}, & \text{otherwise.} \end{cases} \quad (5)$$

A larger τ requires a larger validation-loss reduction for an alternative to be selected over the reference. The margin affects only model selection and does not change the adapted parameters.

The selected candidate provides the architecture and parameters in Eq. (1). Figure 4 illustrates the adaptation and selection stages.

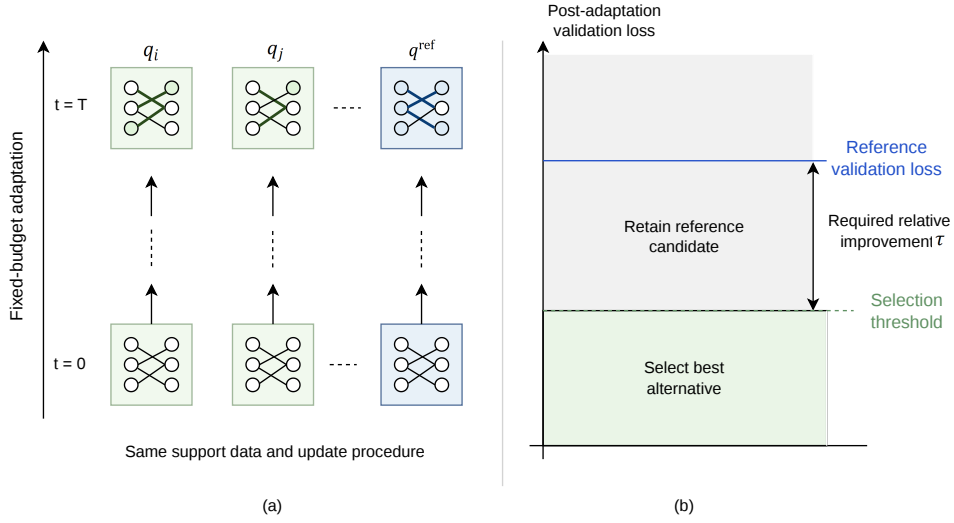


Figure 4: Target adaptation and reference-based model selection.

Algorithm 1 summarizes the target-side procedure.

Algorithm 1 Target-Side MSA-DTI for One Center

Require: Candidate bank $\mathcal{Q}$, reference candidate q^{ref} , support set $\mathcal{D}_c^{\text{sup}}$, validation set $\mathcal{D}_c^{\text{val}}$, deployment limits B_c^F and B_c^P , update number T , step sizes $\{\eta_t\}_{t=0}^{T-1}$, and relative margin τ

Ensure: DT_c or UNSUPPORTEDLIMIT

- 1: Compute $\mathcal{Q}_c^{\text{fea}}$ using Eq. (2)
 - 2: **if** $q^{\text{ref}} \notin \mathcal{Q}_c^{\text{fea}}$ **then**
 - 3: **return** UNSUPPORTEDLIMIT
 - 4: **end if**
 - 5: **for all** $q \in \mathcal{Q}_c^{\text{fea}}$ **do**
 - 6: Adapt q using Eq. (4)
 - 7: Compute $J_c(q)$ using Eq. (3)
 - 8: **end for**
 - 9: $\mathcal{Q}_c^{\text{alt}} \leftarrow \mathcal{Q}_c^{\text{fea}} \setminus \{q^{\text{ref}}\}$
 - 10: **if** $\mathcal{Q}_c^{\text{alt}} = \emptyset$ **then**
 - 11: $q_c^* \leftarrow q^{\text{ref}}$
 - 12: **else**
 - 13: Choose $\tilde{q}_c \in \arg \min_{q \in \mathcal{Q}_c^{\text{alt}}} J_c(q)$ using the deterministic tie rule
 - 14: Select q_c^* using Eq. (5)
 - 15: **end if**
 - 16: $\text{DT}_c \leftarrow (a_{q_c^*}, \theta_{c, q_c^*}^{(T)})$
 - 17: **return** DT_c
-

4.6. Relative-Margin Calibration

The margin τ is calibrated on a margin-calibration pool that is disjoint from the architecture-screening cases and final target evaluation. Each calibration case contains support, validation, and check sets. A total of 20 calibration

centers contribute four cases each, giving 80 cases and 20 independent center clusters for resampling. A selection is harmful when the selected alternative has a higher check-set loss than the adapted reference.

MSA-DTI evaluates preset margins and chooses the smallest value that meets the requirements for harmful rate, average check-loss gain, and statistical confidence. The bank, reference candidate, and selected margin are then fixed before final target evaluation. The exact margin grid and calibration criteria are reported in Section 5.1 and Supplementary Section S2.

The 5% harmful-rate requirement is an ex ante design tolerance chosen to limit erroneous alternative replacement during calibration. It is applied to the observed point estimate rather than as a distribution-free safety guarantee. Eligibility also requires both mean check-loss gains to reach 3% and both lower 95% confidence bounds to remain above zero. If no margin is eligible in a new domain, calibrated deployment returns the adapted reference and reports that domain-specific margin calibration is unsupported. A previously frozen margin may still be evaluated as a transfer audit, but that audit is not reported as calibrated reliability for the new domain.

4.7. Analytical Properties

Deployment feasibility. For every supported target limit setting, MSA-DTI returns a candidate only from Q_c^{fea} ,

$$F_c(a_{q_c^*}) \leq B_c^F, \quad P_c(a_{q_c^*}) \leq B_c^P.$$

Thus, every returned DT model satisfies both deployment limits.

Required validation improvement. When an alternative is selected and $J_c(q^{\text{ref}}) > 0$, Eq. (5) gives

$$\frac{J_c(q^{\text{ref}}) - J_c(q_c^*)}{J_c(q^{\text{ref}})} \geq \tau.$$

Thus, every selected alternative reaches the preset relative improvement.

A conditional expected-loss result and the target-stage computational complexity are provided in Supplementary Section S3.

5. Experimental Evaluation

We evaluate prediction accuracy, selection reliability, deployment feasibility, target-side cost, and robustness across target conditions. Additional component and transfer analyses are also reported.

5.1. Experimental Setup

5.1.1. Benchmark and Data Protocol

We use a controlled synthetic benchmark for heterogeneous multi-center workloads. Centers differ in workload dynamics, sampling intervals, monitored-variable availability, and observation conditions. Deployment-limit tiers are treated as separate platform-side settings. Types A–C are evaluation strata generated from different parameter ranges for these center-side factors. The detailed generation procedure is given in Supplementary Section S8.

The main evaluation contains 20 source centers and 20 held-out target centers. Two prediction horizons and two support sizes give four cases per target center and 80 held-out cases in total.

Each case is split chronologically into support, validation, check, and test sets, with gaps between adjacent splits. The source, architecture-screening, margin-calibration, diagnostic, and held-out target pools use disjoint center identities. Test data are used only after the architecture and adapted parameters have been fixed. Table 1 reports the main settings.

5.1.2. Candidate Bank, Reference, and Calibration

The initial space contains 66 MLP, TCN, and GRU configurations. PT-A57 is the protected pooled-source initialization of a one-layer GRU-32, fixed before screening and shared with PT+FT. Candidate discovery evaluates separately source-trained C1 candidates, including C1-A57. Here C1 denotes the legacy source-pooled initialization bank defined in Section 3; it is not a target-center label. A C1 configuration is retained after at least two screening cases in either of two ways: it is the check-loss winner over PT-A57, or validation selects it and it has positive check-loss

gain over PT-A57. Only support, validation, and check data from the screening pool are used. This gives six retained configurations, five executable architectures, and seven candidates.

Table 1 reports the training and target-adaptation settings, and Supplementary Section S2 provides the candidate-bank construction and calibration details.

Table 1: Main experimental configuration.

Category	Item	Setting
Data	Centers; observations; input length Target cases and data	Source / held-out: 20/20; 3000 per center; $L = 96$ 80 cases; $H \in \{1, 4\}$; $K \in \{10, 20\}$; validation / check / test: 80/60/200
Bank	Initial / retained / executable / candidates Reference and screening Source-model records	66/6/5/7 A57: one-layer GRU-32, fixed before screening and shared with PT+FT; alternative kept after wins in at least two screening cases Reference pooled-source initialization; five common alternative initializations; one additional A57 paired control (Supplementary S2)
Training	Source models Target adaptation	Adam / MSE / 50 epochs; learning rate 10^{-3} ; batch size 64 SGD / MSE / 50 updates; learning rate 0.01; gradient limit 1.0; complete support set
Selection	Grid and eligibility Frozen margin	{0.05, 0.075, 0.10, 0.125, 0.15, 0.20}; harmful rate $\leq 5\%$; both mean gains $\geq 3\%$; both lower 95% bounds > 0 $\tau = 0.10$
Limits	Tight / medium / loose	Operations: $1.5 \times 10^6 / 5 \times 10^6 / 2 \times 10^7$; parameters: $3 \times 10^4 / 10^5 / 5 \times 10^5$
Statistics	Confidence intervals	Center-cluster bootstrap; 4000 repetitions

5.1.3. Task, Baselines, and Metrics

We use workload prediction as the evaluation task because it supports proactive resource management in computing systems [17–19]. Mean squared error (MSE) is the task loss. We also report mean absolute error (MAE), Worst-10% error, conditional value at risk at the 90th percentile (CVaR90), target-side time, and limit satisfaction. Worst-10% is the mean window-level MSE over the 10% of test windows with the largest errors. CVaR90 is the mean case-level MSE over the 10% of target cases with the largest MSE.

For held-out reporting, a selected alternative is labeled harmful when its test MSE is higher than that of the adapted reference. This label is used only after selection. Relative gains are computed for each paired case and then averaged. Confidence intervals use a center-cluster percentile bootstrap with 4000 repetitions, keeping the four cases from one center together. Reported subgroup intervals are unadjusted pointwise 95% intervals.

We compare MSA-DTI with PT+FT, MeDeT-based adaptation, random initialization, Meta+NAS-lite, H-Meta-NAS, zero-shot NAS, and zero-shot NAS with fine-tuning. PT+FT is the main controlled baseline because it uses the same source-trained reference and target adaptation procedure as MSA-DTI. Meta+NAS-lite is a lightweight NAS baseline, and H-Meta-NAS is a hardware-aware meta-NAS baseline. Both search baselines are evaluated under the same 66-architecture space and hard deployment limits. Their complete target-side settings are reported in Supplementary Table S2.

All reported baselines use the same target cases, chronological splits, deployment limits, and architecture definitions. Target-side baseline settings are summarized in Supplementary Table S2, with complete configurations provided in the released implementation.

5.1.4. Runtime Protocol

All methods are implemented in Python and PyTorch and run on an NVIDIA RTX 3060 Laptop GPU. The repeated target-side timing record uses Python 3.9.23, PyTorch 2.5.1+cu121, CUDA 12.1, cuDNN 9.1.0, and float32 model inputs and parameters. Target-side time includes model creation, initialization loading, target adaptation, validation evaluation, and final selection, but excludes offline preparation and test evaluation. Supplementary Section S5 reports the complete timing and hardware-profiling protocol.

5.2. Overall Performance and Selection Reliability

Figure 5 compares prediction performance on the held-out cases. All reported outputs satisfy both deployment limits.

Few-Shot DT Predictive-Model Instantiation



Figure 5: Predictive performance on the held-out target cases: (a) MAE, (b) MSE, (c) Worst-10% error, and (d) CVaR90 (lower is better).

MSA-DTI gives the lowest values for all four error measures. PT+FT is the strongest baseline. Among the reported search baselines, Meta+NAS-lite obtains MSE 0.007105 and H-Meta-NAS obtains MSE 0.013574, compared with 0.005120 for PT+FT and 0.004219 for MSA-DTI. H-Meta-NAS is the hardware-aware meta-NAS baseline. Relative to PT+FT, MSA-DTI reduces MAE by 8.55%, MSE by 14.60%, and Worst-10% error by 13.76% on average across paired cases. The center-cluster bootstrap 95% confidence interval for the paired MSE reduction is [9.41%, 20.14%].

Figure 6 compares MSA-DTI with the adapted reference at the case level. MSA-DTI retains the reference in 33 cases and selects an alternative in 47 cases. Of these alternatives, 44 have lower test MSE and three have higher test MSE. Harmful selections therefore account for 3.75% of all held-out cases. These test labels do not affect selection. The center-cluster bootstrap 95% confidence interval for the all-case harmful-selection rate is [0.00%, 7.50%]; the interval therefore does not establish a population harmful rate below 5%.

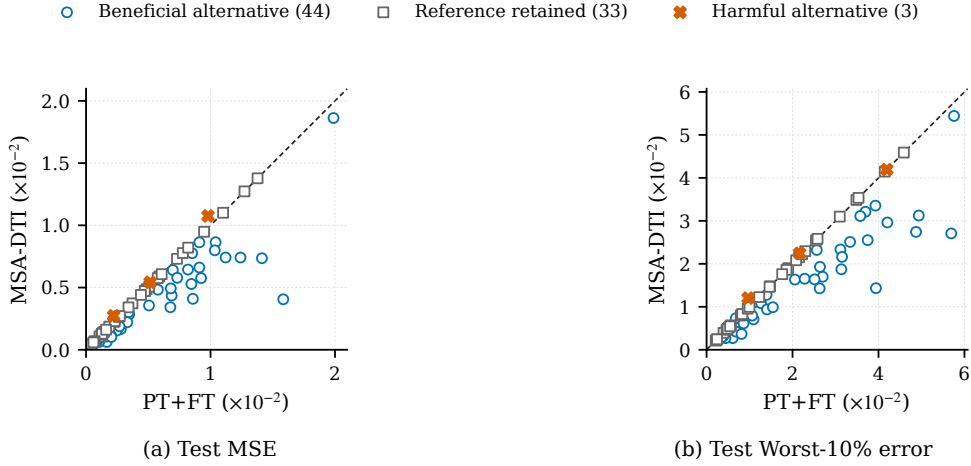


Figure 6: Held-out paired comparison of MSA-DTI and PT+FT for (a) test MSE and (b) test Worst-10% error; points below the diagonal favor MSA-DTI.

5.2.1. Native-protocol comparison

The main comparison reports each baseline under its method-specific native target protocol; Supplementary Table S2 provides the complete settings. This comparison reflects the end-to-end methods as specified, but it does not match all target-side optimizers and losses.

5.2.2. Optimizer-matched diagnostic comparison

Because the native NAS baselines use different target optimizers and losses, the optimizer-matched diagnostic comparison uses SGD, MSE, and 50 target updates for every included method. MSA-DTI keeps the lowest MSE, Worst-10% error, and CVaR90. Table 2 reports the four-method optimizer-matched comparison, and Supplementary Table S3 gives the complete six-method diagnostic.

Table 2: Optimizer-matched diagnostic comparison.

Method	MSE↓	Worst-10%↓	CVaR90↓	Adapted models
PT+FT	0.004497	0.017144	0.011878	1.00
Meta+NAS-lite	0.006715	0.025246	0.019653	12.00
Zero-shot NAS + fine-tuning	0.007402	0.028099	0.020402	12.00
MSA-DTI	0.003809	0.015094	0.008696	6.10

All methods use the same independent diagnostic cases, hard deployment limits, SGD optimizer, MSE loss, and 50 target updates per adapted model. The source initializations and candidate-selection rules remain method-specific. Thus, this diagnostic isolates the effect of the target-side optimizer, loss, and update budget but is not a claim of full algorithmic equivalence. The complete six-method diagnostic is reported in Supplementary Table S3.

5.3. Robustness across Target Conditions

Figure 7 reports paired MSE reductions relative to PT+FT. Positive values favor MSA-DTI.

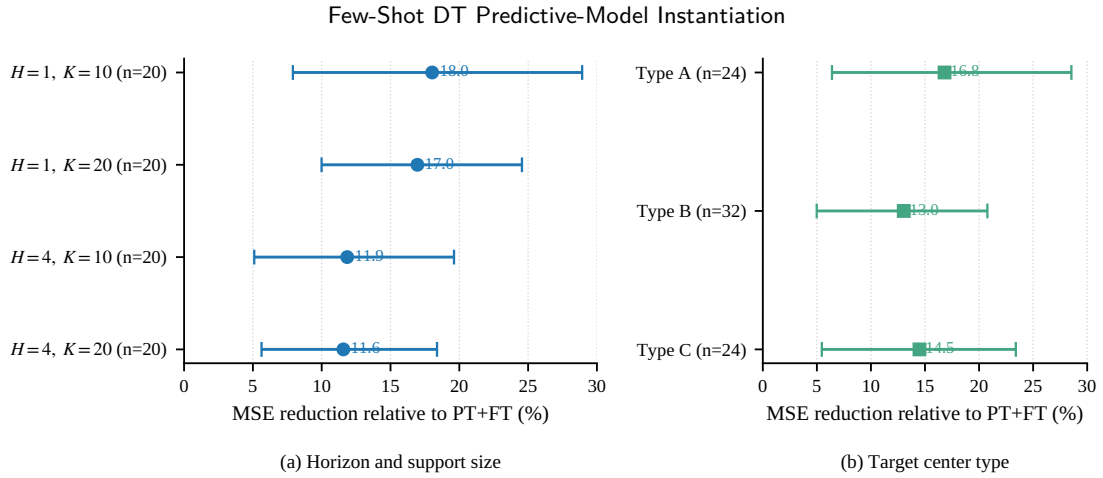


Figure 7: Paired MSE reduction relative to PT+FT across (a) horizon/support settings and (b) target-center types; error bars show 95% confidence intervals.

The reduction ranges from 11.57% to 18.03% across the four horizon and support-size settings and from 13.03% to 16.80% across the three center types. The type panels contain 24, 32, and 24 cases but only 6, 8, and 6 independent center clusters, respectively. Their pointwise intervals are therefore interpreted as exploratory small-cluster summaries rather than formal simultaneous subgroup inference. All seven observed confidence intervals remain above zero.

On a separate robustness pool, retraining the fixed candidate architectures with source-training seeds 2904, 2905, and 2906 gives paired MSE reductions of 10.52%, 13.21%, and 11.25%, respectively. Their 95% confidence intervals are [3.93%, 18.17%], [7.62%, 18.89%], and [5.23%, 17.96%]. This analysis tests source-initialization sensitivity after the architecture list has been fixed.

We additionally repeat the complete 66-configuration screening rule on two new 20-center pools, giving three screening samples when the original pool is included, and evaluate each resulting list on a fourth, untouched 20-center pool. The retained-list sizes are 6, 0, and 4; the two new lists have Jaccard similarities of 0 and 0.429 to the frozen six-configuration list. On the independent pool, validation-only diagnostic selection from the frozen list gives a 0.30% mean paired check-MSE reduction with a center-cluster 95% interval of $[-0.65\%, 1.56\%]$ and a 2.50% harmful-selection rate. Thus, the preset candidate-discovery rule is sample-sensitive and is not claimed to recover a unique architecture set. The frozen main evaluation remains an evaluation of a prespecified bank, while the retained A57 reference and fallback protect deployment when a rescreened pool retains no alternative. Full rescreening results and selection frequencies are reported in Supplementary Section S7.

5.4. Deployment Feasibility and Cost

Figure 8 shows how the three deployment-limit settings affect candidate feasibility and final model selection.

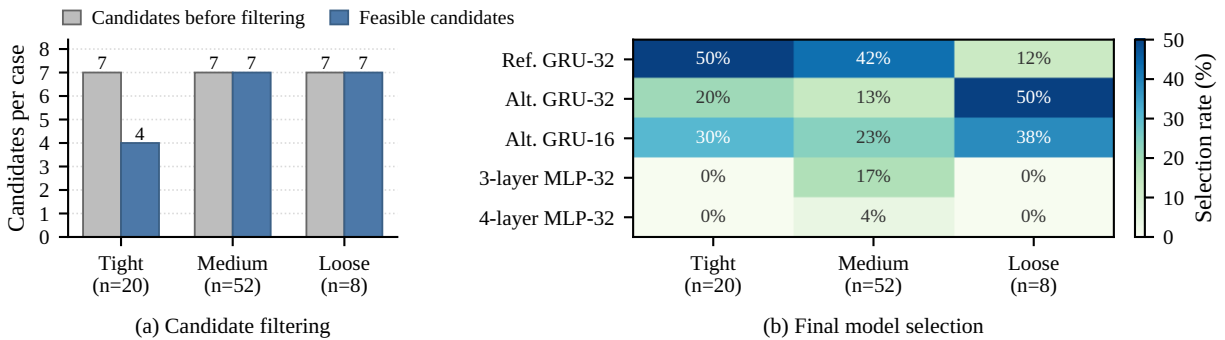


Figure 8: Effects of the three deployment-limit settings on (a) the number of feasible candidates and (b) the selected model configurations.

The tight setting reduces the bank from seven candidates to four, whereas the medium and loose settings retain all seven. A single-limit audit in Supplementary Section S5 shows that this reduction is caused by the parameter-count

limit: all seven candidates satisfy the operation-count limit, while the tight parameter-count limit excludes three MLP candidates. To examine whether the two limits can independently constrain candidate feasibility beyond the frozen main bank, Supplementary Section S5 also reports an expanded-bank stress analysis. For both $H = 1$ and $H = 4$, the operation-count and parameter-count limits exclude different candidates: each single-limit feasible set retains seven of eight candidates, whereas their joint feasible set retains six. The target-side stress evaluation uses the same 80 held-out cases, and all outputs selected under the joint limits satisfy both limits. Mean paired MSE reductions relative to PT+FT remain positive under tight, medium, and loose limits: 11.35%, 15.30%, and 18.22%, respectively. These tier-stratified values are descriptive rather than causal effects of the limits. Tier assignment is type-dependent, and the tight, medium, and loose strata contain only 20, 52, and 8 cases (approximately 5, 13, and 2 independent center clusters), respectively; center type and tier are therefore not independently identified in this benchmark.

Table 3 reports raw target-side cost and selected-model complexity.

Table 3: Target-side cost and selected-model complexity.

Method	Time per case (s)	Adapted models	Parameters	Estimated operations
PT+FT	0.204 ± 0.004	1.00	5,746	1,050,784
Meta+NAS-lite	13.471 ± 0.173	12.00	30,995	1,468,894
Zero-shot NAS + fine-tuning	17.601 ± 0.150	12.00	62,076	1,571,621
MSA-DTI	5.676 ± 0.059	6.25	14,892	751,433

Time is mean $\pm$ standard deviation across five repeat means. Complexity values are averaged over the final models selected in the held-out cases.

MSA-DTI requires 5.676 ± 0.059 seconds per case (where $\pm$ is the standard deviation across the five repeat means) and adapts 6.25 models on average. It is slower than PT+FT, but is 2.37 $\times$ faster than few-shot NAS and 3.10 $\times$ faster than zero-shot NAS with fine-tuning.

Hardware profiling in Supplementary Section S5 reports every executable architecture on an Intel Core i7-12700H CPU and an NVIDIA RTX 3060 Laptop GPU. Median batch-one latency ranges from 0.093 to 1.980 ms on the CPU and from 0.133 to 4.170 ms on the GPU. Serialized model size ranges from 10.3 to 315.9 KiB, and measured peak GPU allocation ranges from 54 to 1037 KiB. Operation count is positively associated with median latency, whereas parameter count exactly orders serialized model size in the measured settings. Peak GPU allocation is not monotonic in parameter count. These measurements support the use of the two counts as distinct complexity indicators, but they do not convert either limit into a hardware latency or memory guarantee or establish cross-platform deployment performance.

5.5. Component Analysis

Table 4 reports the component variants on the separate diagnostic pool. The full method is compared with reference-only, paired-initialization, source-bank, selection, and deployment-filtering controls; this table is not part of the held-out headline comparison.

Table 4: Component ablation on the separate diagnostic pool.

Variant	MSE $\downarrow$	Worst-10% $\downarrow$	Paired MSE reduction (%) $\uparrow$	Harmful rate (%) $\downarrow$	Limit satisfaction (%) $\uparrow$
MSA-DTI	0.004063	0.015764	15.79	15.00	100.00
Earlier source-training procedure	0.004876	0.019044	1.75	2.50	100.00
Reference candidate only	0.005051	0.019512	0.00	0.00	100.00
Two A57 initializations	0.004917	0.019136	1.29	1.25	100.00
Without relative margin	0.004055	0.015690	16.16	17.50	100.00
Without deployment filtering	0.004037	0.015635	16.13	13.75	96.25

Paired MSE reduction uses the adapted reference on the same diagnostic pool. The earlier source-training row changes the alternative-model source-training procedure and is interpreted as source-bank sensitivity rather than an isolated single-component removal.

Removing deployment filtering reduces limit satisfaction from 100% to 96.25%, confirming that filtering is required to enforce the configured hard limits. Direct lowest-validation-loss selection increases the harmful-selection rate from 15.00% to 17.50% on the diagnostic pool, although the confidence intervals overlap. The latter result is therefore an observed difference rather than a statistically confirmed reduction in harmful selection. The reference-only and two-reference-initialization controls in Supplementary Section S4 do not reproduce the cross-architecture gain. Conversely, removing the relative margin leaves the diagnostic MSE nearly unchanged while selecting alternatives more often. The main performance increment is therefore associated with adapting and selecting across the compact architecture bank;

the margin is a conservative replacement rule, not the sole source of the accuracy gain. Supplementary Section S4 provides the corresponding protocol details and additional selection diagnostics.

5.6. External Transfer Evaluation

We also evaluate the frozen method on Alibaba Cluster Trace v2018 [29]. The workload observations come from the production trace, while the deployment-limit tiers remain controlled model-complexity settings. This experiment therefore evaluates external workload-domain transfer rather than a fully measured hardware-deployment setting.

Without recalibrating the margin, the fixed $\tau = 0.10$ rule gives a 4.20% mean paired MSE reduction with a 95% confidence interval of [2.03%, 6.70%]. The harmful-selection rate is 6.25%, which exceeds the original 5% criterion. The result therefore shows a positive average transfer gain but does not reproduce the original reliability criterion. Under the calibrated deployment policy defined in Section 4, this outcome retains the adapted reference rather than treating the frozen margin as certified for Alibaba; the reported frozen-margin result is a transfer audit. The full transfer protocol and results are given in Supplementary Section S7.

6. Conclusion

This paper presented MSA-DTI for few-shot instantiation of the predictive-model component used by a center-specific DT, subject to two per-model deployment limits. MSA-DTI filters a source-trained candidate bank, adapts feasible candidates with a common target budget, and applies reference-based selection. MSA-DTI reduces MSE by 14.60% on average across paired cases relative to PT+FT while satisfying both limits, and 44 of 47 alternative selections improve test MSE. Independent rescreening shows that candidate discovery is sample-sensitive, so the frozen bank is treated as a prespecified evaluated artifact rather than a uniquely recoverable architecture set; the reference fallback remains essential. The transferred rule also gives a positive mean gain on Alibaba workloads, although its harmful-selection rate exceeds the original criterion and the calibrated deployment policy therefore retains the reference. Future work will develop more stable candidate-discovery procedures, uncertainty-aware selection, and measured deployment limits on additional platforms.

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The final author contribution statement will be added after the author list is confirmed.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

The source code, frozen configurations, processed results, plot-ready data, and reproduction scripts underlying the reported results are available in GitHub Release v1.2.3: <https://github.com/w924871681/paper-code-DT-Twin/releases/tag/v1.2.3>. The release includes the complete reproduction package `rcf_dti_v1.2.3_complete.zip`, the paper-alignment package `paper_alignment_v1.2.3.zip`, the bootstrap-analysis package `level_c_bootstrap_v1.2.3.zip`, and the CUDA replay evidence `cuda_replay_evidence_v1.2.3.zip`. The original Alibaba Cluster Trace v2018 is not redistributed and remains available from its official repository: <https://github.com/alibaba/clusterdata>. The released documentation specifies preprocessing, disjoint data splits, margin handling, and held-out evaluation.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During the preparation of this work, the authors used OpenAI ChatGPT to assist with language refinement, manuscript organization, and readability improvement. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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